Chapter 9

Composite Likelihood

Composite likelihoods appear to have originated with Lindsay (1988). This idea attracted a great deal of attention in the early 2000's, and remains an area of active research today. Overviews of this topic can be found in Varin (2008) and Varin, Reid and Firth (2011).

9.1 Definition

Consider an $m-$dimensional random vector $\boldsymbol{Y} = (Y_1, \ldots, Y_m)^T$ that has a joint probability density or mass function $f(\boldsymbol{y}|\boldsymbol{\theta})$ for some parameter $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)^T \in \Theta$. The random vector $\boldsymbol{Y}$ may have component variables corresponding to observation of the same quantity on m sampling units or may have component variables corresponding to observations of different quantities on the same sampling unit. Situations in which composite likelihoods are valuable typically involve models that specify dependence structures among the component variables of $\boldsymbol{Y}$. For example, $\boldsymbol{Y}$ may correspond to observation of a single quantity at m locations in a geographical region (e.g., field,

county, or state), or observation of a single quantity at m time points for an individual patient in a clinical trial. Alternatively, $\boldsymbol{Y}$ may correspond to observations of m different attributes of a sampling unit, such as total income, total debt, and assessed value for households. Even more complex situations are possible to consider, such as observation of income, debt, and assessed value for households at given geographic locations in a city, county, or state.

Given a random vector $\boldsymbol{Y}$, consider a set of marginal and/or conditional events $\{\mathcal{A}_k : k = 1, \ldots, K\}$. For example, if the components of $\boldsymbol{Y}$ are discrete, a given event $\mathcal{A}_k$ might be the event that $Y_1 = y$ or the event $(Y_1 = y_1) \cap (Y_2 = y_2)$ or the event that $Y_1 = y_1$ given that $Y_2 = y_2$, or even the event that $(Y_1 = y_1) \cap (Y_2 = y_2) \cap \ldots \cap (Y_m = y_m)$. Let $L_k(\boldsymbol{\theta}\boldsymbol{y})$ denote the likelihood of event $\mathcal{A}_k$ that is dictated by the joint pdf or pmf $f(\boldsymbol{y}|\boldsymbol{\theta})$. A composite likelihood is then defined as

$$L_c(\boldsymbol{\theta}|\boldsymbol{y}) = \prod_{k=1}^{K} [L_k(\boldsymbol{\theta}|\boldsymbol{y})]^{w_k}, \tag{9.1}$$

where w_k may be a non-negative weight for $k = 1, \ldots, K$. The log composite likelihood is then

$$\ell_c(\boldsymbol{\theta}|\boldsymbol{y}) = \log\{L_c(\boldsymbol{\theta}|\boldsymbol{y})\} = \sum_{k=1}^{K} w_k \, \log\{L_k(\boldsymbol{\theta}|\boldsymbol{y})\} = \sum_{k=1}^{K} w_k \, \ell_k(\boldsymbol{\theta}|\boldsymbol{y}). \tag{9.2}$$

We will assume for the remainder of this chapter that the weights w_k are all the same and may be set identically equal to 1. A value $\hat{\boldsymbol{\theta}}$ is a maximum composite likelihood estimate if, for all $\boldsymbol{\theta} \in \Theta$,

$$L_c(\boldsymbol{\theta}|\boldsymbol{y}) \leq L_c(\hat{\boldsymbol{\theta}}|\boldsymbol{y}) \quad \text{or} \quad \ell_c(\boldsymbol{\theta}|\boldsymbol{y}) \leq \ell_c(\hat{\boldsymbol{\theta}}|\boldsymbol{y}). \tag{9.3}$$

Clarification of what structures qualify as composite likelihoods is perhaps best obtained through examples. Typically, the events $\{\mathcal{A}_k : k = 1, \ldots, K\}$

correspond to either all marginal or all conditional events, although mixtures of marginal and conditional events would be possible.

1. Independence Likelihood

   What is often called an independence likelihood results from using k to index individual elements of $\boldsymbol{Y}$ so that $K = m$, $\mathcal{A}_k$ is the event that $Y_k = y_k$ (or $y_k - \delta < Y_k < y_k + \delta$ in the continuous case) and

$$L_c(\boldsymbol{\theta}|\boldsymbol{y}) = \prod_{k=1}^{m} f(y_k|\boldsymbol{\theta}), \tag{9.4}$$

   where $f(y_i|\boldsymbol{\theta})$ is the marginal pdf or pmf of Y_i dictated by the model $f(\boldsymbol{y}|\boldsymbol{\theta})$. This is called an independence (composite) likelihood for fairly obvious reasons – it is the likelihood we would construct if the model assumed the elements of $\boldsymbol{Y}$ were independent.

2. Pairwise Marginal Likelihood

   A pairwise marginal likelihood is constructed by taking k to index distinct pairs of elements of $\boldsymbol{Y}$ so that $K = m(m-1)/2$, $\mathcal{A}_k$ is the event that $(Y_i = y_i) \cap (Y_j = y_j)$; $(i, j) = k$ (or its continuous analog), and

$$\begin{aligned} L_c(\boldsymbol{\theta}|\boldsymbol{y}) &= \prod_{i=1}^{m-1} \prod_{j=i+1}^{m} f(y_i, y_j|\boldsymbol{\theta}) \\ &= \prod_{1 \le i < j \le m} f(y_i, y_j|\boldsymbol{\theta}), \end{aligned} \tag{9.5}$$

   where $f(y_i, y_j|\boldsymbol{\theta})$ is the bivariate marginal pdf or pmf dictated for Y_i and Y_j by the joint model $f(\boldsymbol{y}|\boldsymbol{\theta})$. It should be clear that such a pairwise marginal likelihood could be extended to be a composite likelihood formed from triples or quadruples or margins of any dimension.

3. Pairwise Conditional Likelihood

   A pairwise conditional likelihood is constructed by taking k to index all pairs of elements of $\boldsymbol{Y}$ so that $K = m(m-1)$, $\{\mathcal{A}_k$ is the event that $Y_i = y_i$ given that $Y_j = y_j$ for $(i,j) = k$, and

$$L_c(\boldsymbol{\theta}|\boldsymbol{y}) = \prod_{i=1}^{n} \prod_{j \neq i} f(y_i|y_j, \boldsymbol{\theta}), \tag{9.6}$$

   where $f(y_i|y_j, \boldsymbol{\theta})$ are conditional pdfs or pmfs dictated by $f(\boldsymbol{y}|\boldsymbol{\theta})$.

4. Pairwise Difference Likelihood

   A pairwise difference likelihood is constructed by taking k to index distinct pairs of elements of $\boldsymbol{Y}$ so that $K = m(m-1)/2$, $\mathcal{A}_k$ is the event that $(Y_i - Y_j) = (y_i - y_j)$; $(i,j) = k$, and

$$\begin{aligned} L_c(\boldsymbol{\theta}|\boldsymbol{y}) &= \prod_{i=1}^{m-1} \prod_{j=i+1}^{m} f(y_i - y_j|\boldsymbol{\theta}) \\ &= \prod_{1 \leq i < j \leq m} f(y_i - y_j|\boldsymbol{\theta}), \end{aligned} \tag{9.7}$$

   where $f(y_i, -y_j|\boldsymbol{\theta})$ is the pdf or pmf dictated for $(Y_i - Y_j)$ by the joint model $f(\boldsymbol{y}|\boldsymbol{\theta})$.

5. Besag's Original Pseudo-likelihood

   In the context of spatial problems formulated on a discrete-index random field, Besag (1975) suggested the use of what he called a pseudo-likelihood function for which k indexes individual elements of $\boldsymbol{Y}$ so that $K = m$, $\mathcal{A}_k$ is the event that $Y_k = y_k$ given values for $\{Y_j : j \neq k\}$ and

$$L_c(\boldsymbol{\theta}|\boldsymbol{y}) = \prod_{k=1}^{m} f(y_k|\{y_j : j \neq k\}, \boldsymbol{\theta}), \tag{9.8}$$

   where $f(y_k|\{y_j : j \neq k\})$ is the full conditional pdf or pmf dictated by $f(\boldsymbol{y}|\boldsymbol{\theta})$.

9.2 Composite Scores as Estimating Functions

Beginning with $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)^T$ and $w_k = 1$ for all k in (9.2), define the composite score function as

$$U_c(\boldsymbol{\theta}|\boldsymbol{y}) = (U_1(\boldsymbol{\theta}|\boldsymbol{y}), \ldots, U_p(\boldsymbol{\theta}|\boldsymbol{y}))^T, \tag{9.9}$$

where, for $j = 1, \ldots, p$,

$$U_j(\boldsymbol{\theta}|\boldsymbol{y}) = \frac{\partial}{\partial \theta_j} \ell_c(\boldsymbol{\theta}|\boldsymbol{y}) = \sum_{k=1}^{K} \frac{\partial}{\partial \theta_j} \ell_k(\boldsymbol{\theta}|\boldsymbol{y}). \tag{9.10}$$

It will be useful in what follows to note that $U_c(\boldsymbol{\theta}|\boldsymbol{y})$ may also be written as,

$$U_c(\boldsymbol{\theta}|\boldsymbol{y}) = \sum_{k=1}^{K} U_k(\boldsymbol{\theta}|\boldsymbol{y}), \tag{9.11}$$

where,

$$U_k(\boldsymbol{\theta}|\boldsymbol{y}) = \left(\frac{\partial}{\partial \theta_1} \ell_k(\boldsymbol{\theta}|\boldsymbol{y}), \ldots, \frac{\partial}{\partial \theta_p} \ell_k(\boldsymbol{\theta}|\boldsymbol{y}) \right)^T. \tag{9.12}$$

The components of a composite likelihood are true likelihood "pieces" so that, as long as we can interchange differentiation and integration,

$$E\{U_c(\boldsymbol{\theta}|\boldsymbol{y})\} = \mathbf{0}. \tag{9.13}$$

Verifying that this is the case is not entirely trivial. Expression (9.1) is a general definition of a composite likelihood in which $L_k(\boldsymbol{\theta}|\boldsymbol{y})$ is either a marginal or conditional distribution dictated by the joint model $f(\boldsymbol{y}|\boldsymbol{\theta})$. Consider the representation of $U_c(\boldsymbol{\theta}|\boldsymbol{y})$ given in (9.9) and (9.10). Suppose first that $L_k(\boldsymbol{\theta}|\boldsymbol{y}) = f(\boldsymbol{y}_s|\boldsymbol{\theta})$ where $f(\boldsymbol{y}_s|\boldsymbol{\theta})$ is the joint marginal pdf or pmf for a subset of random variables $\boldsymbol{Y}_s \subset \boldsymbol{Y}$. Then, for any $j = 1, \ldots, p$ and dominating

measure μ,

$$
\begin{aligned}
E\{U_j(\boldsymbol{\theta}|\boldsymbol{y})\} &= E\left[\sum_{k=1}^{K} \frac{\partial}{\partial\theta_j} \log\{L_k(\boldsymbol{\theta}|\boldsymbol{y})\}\right] \\
&= \sum_{k=1}^{K} E\left[\frac{\partial}{\partial\theta_j} \log\{f(\boldsymbol{y}_s|\boldsymbol{\theta})\}\right] \\
&= \sum_{k=1}^{K} E\left[\frac{1}{f(\boldsymbol{y}_s|\boldsymbol{\theta})}\left\{\frac{\partial}{\partial\theta_j} f(\boldsymbol{y}_s|\boldsymbol{\theta})\right\}\right] \\
&= \sum_{k=1}^{K} \int_{-\infty}^{\infty} \frac{1}{f(\boldsymbol{y}_s|\boldsymbol{\theta})}\left\{\frac{\partial}{\partial\theta_j} f(\boldsymbol{y}_s|\boldsymbol{\theta})\right\} f(\boldsymbol{y}|\boldsymbol{\theta})\, d\mu(\boldsymbol{y}) \\
&= \sum_{k=1}^{K} \int_{-\infty}^{\infty} \frac{1}{f(\boldsymbol{y}_s|\boldsymbol{\theta})}\left\{\frac{\partial}{\partial\theta_j} f(\boldsymbol{y}_s|\boldsymbol{\theta})\right\} f(\boldsymbol{y}_s|\boldsymbol{\theta})\, d\mu(\boldsymbol{y}_s) \\
&= \sum_{k=1}^{K} \int_{-\infty}^{\infty} \frac{\partial}{\partial\theta_j} f(\boldsymbol{y}_s|\boldsymbol{\theta})\, d\mu(\boldsymbol{y}_s) \\
&= \sum_{k=1}^{K} \frac{\partial}{\partial\theta_j} \int_{-\infty}^{\infty} f(\boldsymbol{y}_s|\boldsymbol{\theta})\, d\mu(\boldsymbol{y}_s) = 0. \qquad (9.14)
\end{aligned}
$$

Now suppose that $L_k(\boldsymbol{\theta}|\boldsymbol{y}) = f(\boldsymbol{y}_s|\boldsymbol{y}_r, \boldsymbol{\theta})$, where $f(\boldsymbol{y}_s|\boldsymbol{y}_r, \boldsymbol{\theta})$ is the joint conditional pdf or pmf for $\boldsymbol{Y}_s \subset \boldsymbol{Y}$ given $\boldsymbol{Y}_r \subset \boldsymbol{Y}$. Then, for any $j = 1, \ldots, p$,

$$
\begin{aligned}
E\{U_j(\boldsymbol{\theta}|\boldsymbol{y})\} &= E\left[\sum_{k=1}^{K} \frac{\partial}{\partial\theta_j} \log\{L_k(\boldsymbol{\theta}|\boldsymbol{y})\}\right] \\
&= \sum_{k=1}^{K} E\left[\frac{\partial}{\partial\theta_j} \log\{f(\boldsymbol{y}_s, \boldsymbol{y}_r|\boldsymbol{\theta})\} - \frac{\partial}{\partial\theta_j} \log\{f(\boldsymbol{y}_r|\boldsymbol{\theta})\}\right]. \qquad (9.15)
\end{aligned}
$$

Applying (9.14) to each term in (9.15) (starting at the second line of 9.14) shows that these expectations are also equal to zero. The result is that composite likelihoods have the form of unbiased estimating functions (e.g., Godambe 1991).

We know that for true likelihood functions $L(\boldsymbol{\theta}|\boldsymbol{y})$ and $\ell(\boldsymbol{\theta}|\boldsymbol{y}) = \log\{L(\boldsymbol{\theta}|\boldsymbol{y})\}$,

and any $u, v = 1, \ldots, p$,

$$E\left[\frac{\partial \ell(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_u}\frac{\partial \ell(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_v}\right] = -E\left[\frac{\partial^2 \ell(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_u\, \partial \theta_v}\right]. \tag{9.16}$$

as long as dominated convergence holds. In a manner similar to that given above in expressions (9.14) and (9.15), this equality will also hold for each component likelihood piece of a log composite likelihood, $\ell_k(\boldsymbol{\theta}|\boldsymbol{y})$,

$$E\left[\frac{\partial \ell_k(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_u}\frac{\partial \ell_k(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_v}\right] = -E\left[\frac{\partial^2 \ell_k(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_u\, \partial \theta_v}\right], \tag{9.17}$$

but it does not hold in general for the entire log composite likelihood $\ell_C(\boldsymbol{\theta}|\boldsymbol{y})$. An important quantity for composite likelihoods is what is often called Godambe Information, defined as

$$G(\boldsymbol{\theta}) = H(\boldsymbol{\theta})J^{-1}(\boldsymbol{\theta})H(\boldsymbol{\theta}). \tag{9.18}$$

In (9.18) $J(\boldsymbol{\theta})$ is a $p \times p$ matrix with uv^{th} element

$$J_{u,v}(\boldsymbol{\theta}) = E\left[\left\{\sum_{k=1}^{K}\frac{\partial \ell_k(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_u}\right\}\left\{\sum_{k=1}^{K}\frac{\partial \ell_k(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_v}\right\}\right], \tag{9.19}$$

and $H(\boldsymbol{\theta})$ is a $p \times p$ matrix with uv^{th} element

$$H(\boldsymbol{\theta}) = E\left[-\sum_{k=1}^{K}\frac{\partial^2 \ell_k(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_u\, \partial \theta_v}\right]. \tag{9.20}$$

In much of the literature on composite likelihoods these matrices are written as $J(\boldsymbol{\theta}) = E[U_C(\boldsymbol{\theta}|\boldsymbol{y})\, U_C^T(\boldsymbol{\theta}|\boldsymbol{y})]$ and $H(\boldsymbol{\theta}) = E[-\nabla U_C(\boldsymbol{\theta}|\boldsymbol{y})]$, where ∇ is the gradient operator. Notice that $J(\boldsymbol{\theta})$ gives the variance of $\ell_C(\boldsymbol{\theta})$ while $H(\boldsymbol{\theta})$ is related to the curvature of the log composite likelihood surface. For $\ell_C(\boldsymbol{\theta})$ a full likelihood these are the same, $H(\boldsymbol{\theta}) = J(\boldsymbol{\theta})$ and (9.18) is the usual the Fisher information.

9.3 Composite Likelihood Asymptotics

We have emphasized several times the importance of asymptotic context, and that is certainly true in the case of results for maximum composite likelihood estimators. Some relevant results are sketched here for two asymptotic contexts in which composite likelihoods see use.

9.3.1 Asymptotic Context I

Here, let $\boldsymbol{Y}_i = (Y_{i,1}, \ldots, Y_{i,m})^T$ denote one random variable of the type discussed up to this point, and assume that independent and identically distributed copies of this variables will be available as $\{\boldsymbol{Y}_i : i = 1, \ldots\}$. Let the common joint pdfs or pmfs of the Y_i be $f(\boldsymbol{y}|\boldsymbol{\theta})$ where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)^T$ is a $p-$dimensional parameter, $\boldsymbol{\theta} \in \Theta$. Continuing to take weights w_k in (9.1) to all have the value 1, the composite and log composite likelihoods based on n random variables are

$$\begin{aligned} L_C^{(n)}(\boldsymbol{\theta}|\boldsymbol{y}) &= \prod_{i=1}^{n} L_C(\boldsymbol{\theta}|\boldsymbol{y}_i) = \prod_{i=1}^{n}\prod_{k=1}^{K} L_k(\boldsymbol{\theta}|\boldsymbol{y}_i) \\ \ell_C^{(n)}(\boldsymbol{\theta}|\boldsymbol{y}) &= \sum_{i=1}^{n} \log\{L_C(\boldsymbol{\theta}|\boldsymbol{y}_i)\} = \sum_{i=1}^{n}\sum_{k=1}^{K} \log\{L_k(\boldsymbol{\theta}|\boldsymbol{y}_i)\}. \end{aligned} \tag{9.21}$$

Let $\hat{\boldsymbol{\theta}}_C^{(n)}$ denote the parameter value that maximizes $L_C^{(n)}(\boldsymbol{\theta}|\boldsymbol{y})$ or $\ell_C^{(n)}(\boldsymbol{y})$. In this first asymptotic context we suppose that m remains fixed as $n \to \infty$, so we are collecting more and more copies of the $m-$dimensional random variables. Under regularity conditions similar to those used in determining asymptotic results for full maximum likelihood estimators, we have that

$$\sqrt{n}\,[G(\boldsymbol{\theta})]^{1/2}\,(\hat{\boldsymbol{\theta}}_C^{(n)} - \boldsymbol{\theta}) \xrightarrow{\mathcal{L}} MVN(\mathbf{0},\, I_{p\times p}), \tag{9.22}$$

where $I_{p\times p}$ is the $p \times p$ identity matrix and $G(\boldsymbol{\theta})$ is given in (9.18). Notice here that $G(\boldsymbol{\theta})$ is a non-stochastic matrix that does not depend on n (the sums in (9.19) and (9.20) are over components of the composite likelihood for a single observation, not over replicate observations). We may then consider replacing $G(\boldsymbol{\theta})$ with a consistent estimator – actually, we will use estimators for $J(\boldsymbol{\theta})$ and $H(\boldsymbol{\theta})$ separately.

A sample version of $H(\boldsymbol{\theta})$ is

$$\hat{H}_n(\boldsymbol{\theta}) = -\frac{1}{n}\nabla U_C(\boldsymbol{\theta}|\boldsymbol{y})\Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_C^{(n)}}, \tag{9.23}$$

which means that $\hat{H}_n(\boldsymbol{\theta})$ is $p \times p$ with uv^{th} element

$$h_{u,v} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}\left[\frac{\partial^2 \ell_k(\boldsymbol{\theta}|\boldsymbol{y})}{\partial\theta_u\,\partial\theta_v}\right]\Bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_C^{(n)}}. \tag{9.24}$$

Another possible replacement for $H(\boldsymbol{\theta})$ arises because (9.17) does hold for each component piece of a composite likelihood, and we could take the elements of $\hat{H}_n(\boldsymbol{\theta})$ to be given by

$$h_{u,v} = \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}\left[\frac{\partial \ell_k(\boldsymbol{\theta}|\boldsymbol{y}_i)}{\partial\theta_u}\,\frac{\partial \ell(\boldsymbol{\theta}|\boldsymbol{y}_i)}{\partial\theta_v}\right]\Bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_C^{(n)}}. \tag{9.25}$$

A sample version of $J(\boldsymbol{\theta})$ is

$$\hat{J}_n(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n}\left[U_C(\boldsymbol{\theta}|\boldsymbol{y}_i)\,U_C^T(\boldsymbol{\theta}|\boldsymbol{y}_i)\right]\Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_C^{(n)}}, \tag{9.26}$$

so that $\hat{J}_n(\boldsymbol{\theta})$ is a $p \times p$ matrix with uv^{th} element

$$j_{u,v} = \frac{1}{n}\sum_{i=1}^{n}\left[\sum_{k=1}^{K}\frac{\partial \ell_k(\boldsymbol{\theta}|\boldsymbol{y}_i)}{\partial\theta_u}\,\sum_{k=1}^{K}\frac{\partial \ell_k(\boldsymbol{\theta}|\boldsymbol{y}_i)}{\partial\theta_v}\right]\Bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_C^{(n)}}. \tag{9.27}$$

There are several potential difficulties with the use of sample versions of $H(\boldsymbol{\theta})$ and $J(\boldsymbol{\theta})$ in practice. First, how large n needs to be relative to p in

order for (9.22) to provide a good approximation when sample versions of the Godambe information are used is not easily determined. In addition, and importantly, even though a composite likelihood may have a simpler form than a full likelihood, derivatives needed to determine computational versions of $H(\boldsymbol{\theta})$ and $J(\boldsymbol{\theta})$ can be quite complex, leaving computation a still challenging task. One alternative to the use of sample versions of $G(\theta)$ is to estimate the covariance matrix of $\hat{\boldsymbol{\theta}}_C^{(n)}$ through the use of a jackknife. Let $\hat{\boldsymbol{\theta}}$ denote a maximum composite likelihood estimate for a given set of data $\boldsymbol{y} = \{\boldsymbol{y}_i : i = 1, \ldots, n\}$. Let $\boldsymbol{y}_{-i} = \{\boldsymbol{y}_j : j \neq i\}$ be the data with the ith observation removed and $\hat{\boldsymbol{\theta}}_{-i}$ the composite likelihood based on $\boldsymbol{y}_{-i}$. A jackknife estimate of the covariance of the limiting normal distribution is

$$\hat{C}(\hat{\theta}) = \frac{n-1}{n} \sum_{i=1}^{n} \left(\hat{\boldsymbol{\theta}}_{-i} - \hat{\boldsymbol{\theta}}\right) \left(\hat{\boldsymbol{\theta}}_{-i} - \hat{\boldsymbol{\theta}}\right)^T \tag{9.28}$$

Another alternative that is common in applications is to produce interval estimates directly through the use of parametric bootstrap, rather than rely on asymptotic normality of the maximum composite likelihood estimator with some estimated or computed covariance matrix. Part of the motivation for use of composite likelihoods is usually computational simplicity in the first place, so producing sets of bootstrap estimates is frequently not a major problem.

9.3.2 Asymptotic Context II

In the second asymptotic context we will assume that we have only one random vector $\boldsymbol{Y} = (Y_1, \ldots, Y_m)^T$ with joint pdf or pmf $f(\boldsymbol{y}|\boldsymbol{\theta})$ where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)^T$ is a $p-$dimensional parameter, $\boldsymbol{\theta} \in \Theta$. Continuing to take weights w_k in (9.1) to all have the value 1, the composite and log composite

likelihoods are

$$\begin{aligned} L_C^{(m)}(\boldsymbol{\theta}|\boldsymbol{y}) &= \prod_{k=1}^{K} L_k(\boldsymbol{\theta}|\boldsymbol{y}) \\ \ell_C^{(m)}(\boldsymbol{\theta}|\boldsymbol{y}) &= \sum_{k=1}^{K} \log\{L_k(\boldsymbol{\theta}|\boldsymbol{y})\}. \end{aligned} \tag{9.29}$$

Let $\hat{\boldsymbol{\theta}}_C^{(m)}$ denote the parameter value that maximizes $L_C^{(m)}(\boldsymbol{\theta}|\boldsymbol{y})$ or $\ell_C^{(n)}(\boldsymbol{y})$ and consider $m \to \infty$. Replicate observations of $\boldsymbol{Y}$ are not available, and asymptotic results are more difficult to arrive at that for Asymptotic Context I. Essentially what is needed are restrictions on the model that produce internal replications of the same statistical behaviors, such as are produced by assumptions of stationarity. The two fundamental problems that arise are (1) determining the existence of a sequence of normalizing matrices $\mathcal{I}_m$ such that, as $m \to \infty$, $[\mathcal{I}_m]^{-1/2}(\hat{\boldsymbol{\theta}}_C^{(m)} - \boldsymbol{\theta})$ converges in distribution to a multivariate normal distribution with mean vector $\mathbf{0}$ and covariance given by the $p-$dimensional identity matrix $I_{p\times p}$, and (2) determining how to compute $\mathcal{I}_m$.

There are some general results available for certain groups of regularity conditions, but in general the situation is more fragmented than previously, that is, there is less common structure to be exploited and hence a much greater variety of results under a greater variety of conditions. We will give here one example, that being for a composite likelihood given as Besag's original pseudo-likelihood estimator. Let $\boldsymbol{y}_{-i}$ denote the variables $\{y_j : j \neq i\}$ (don't get the index i here confused with the use of i in the previous section – here, i denotes a particular component of the $m-$vector $\boldsymbol{y}$. Besag's log pseudo-likelihood can be defined as

$$\ell_C^{(m)} = \sum_{i=1}^{m} \log\{f(y_i|\boldsymbol{y}_{-i}, \boldsymbol{\theta})\} = \sum_{i=1}^{m} \ell_i(\boldsymbol{\theta}|\boldsymbol{y}). \tag{9.30}$$

Comets and Janzura (1998) present asymptotic normality of maximum pseudo-likelihood estimators for a spatial setting on a regular lattice. Here, let $N_i;\ i = 1, m$ denote the neighborhoods of the m components of $\boldsymbol{Y}$. Impose a condition of "translation-invariant" conditional specification, but without placing conditions on joint distributions or the strength of statistical dependence associated with a model (more on this to follow). Notationally, let $\ell_i(\boldsymbol{\theta}|\boldsymbol{y}) = \log\{f(y_i|\boldsymbol{y}(N_i))$ be the ith term of Besag's original log pseudo-likelihood (9.30). With all vectors being considered column vectors and the superscript T denoting transpose, the gradient of the log pseudo-likelihood can be written as an element-wise sum of vectors as,

$$U_n^T = \sum_{i=1}^{n} U_i^T = \sum_{i=1}^{n} \left(\frac{\partial \ell_i(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_1}, \ldots, \frac{\partial \ell_i(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_p} \right). \tag{9.31}$$

Let $H_i(\boldsymbol{\theta}, \boldsymbol{Y})$ denote the $p \times p$ matrix of negative second derivatives of $\ell_i(\boldsymbol{\theta}|\boldsymbol{y})$ with uv^{th} element

$$h_{u,v} = -\frac{\partial^2 \ell_i(\boldsymbol{\theta}|\boldsymbol{y})}{\partial \theta_u \, \partial \theta_v}; \quad u, v = 1, \ldots, p. \tag{9.32}$$

Note that, in contrast to the previous section, the elements of H_i are not expected values. The result of Comets and Janzura (see also Gaetan and Guyon, 2010, Theorem 5.5) may now be stated as follows.

<u>Result (Comets and Janzura)</u>

Suppose that $\boldsymbol{Y}$ is a Markov random field on a regular integer lattice with translation-invariant conditional specification and let N_i denote the neighborhood of Y_i. If $\hat{\boldsymbol{\theta}}$ is the original maximum pseudo-likelihood estimator of $\boldsymbol{\theta}$, then

$$[\mathcal{J}_m(\boldsymbol{\theta}, \boldsymbol{Y})]^{-1/2} \, H_m(\boldsymbol{\theta}, \boldsymbol{Y})[\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}] \xrightarrow{\mathcal{L}} \text{MVN}\ (\mathbf{0}, I_{p \times p}), \tag{9.33}$$

where $H_m(\boldsymbol{\theta}, \boldsymbol{Y}) = \sum_{i=1}^{m} H_i(\boldsymbol{\theta}, \boldsymbol{Y})$ with these terms as given in (9.32), and

$\mathcal{J}_m(\boldsymbol{\theta}, \boldsymbol{Y})$ is

$$\mathcal{J}_m(\boldsymbol{\theta}, \boldsymbol{Y}) = \sum_{i=1}^{m} U_i \sum_{j \in \{i \cup N_i\}} U_j, \tag{9.34}$$

with U_i as in (9.31).

There are two aspects of this result that make it particularly worthy of note. First is the condition of translation-invariant conditional specification, which does not involve conditions on the joint distribution or, notably, the strength of dependence, and this is in contrast to most asymptotic results for pseudo-likelihood estimation. For a formal statement of this condition see Gaetan and Guyon (2010, pp. 63-64). The condition of translation-invariance for conditional specifications is, however, rather strong in its own way. Essentially, translation-invariance of a conditional specification implies that the conditional distributions specified in the model are identical for each location. Note that this eliminates non-constant mean structure in space, or dependence that varies over time for a lattice that includes locations defined in both space and time. A second aspect of the Result that is interesting is the form of the inner summation in (9.34). An algebraic identity gives that

$$U^T U = \sum_{i=1}^{m} U_i \sum_{j=1}^{m} U_j, \tag{9.35}$$

and $U^T U$ or its average $(1/m)\, U^T U$ appears in any number of asymptotic results in the literature; in a straight likelihood problem, for example, $(1/m)\, U^T U$ is one possible estimator of Fisher information. In (9.34), however, the inner summation is over only the union of each site with its neighbors, not over all sites. To my knowledge, this makes the result of Comets and Janzura unique in the identification of sequences of normalizing matrices $\mathcal{I}_m$. As described previously, this result implies that, if m is sufficiently large, we may behave as if $\hat{\boldsymbol{\theta}}_C^{(m)}$ has a normal distribution with covariance

matrix $\mathcal{I}_m^{-1} = H_m^{-1}(\boldsymbol{\theta}, \boldsymbol{Y})\, \mathcal{J}_m(\boldsymbol{\theta}, \boldsymbol{Y})\, H_m^{-1}(\boldsymbol{\theta}, \boldsymbol{Y})$. The matrices J_m and H_m can be estimated by evaluating them at the maximum pseudo-likelihood (or composite likelihood) estimator.

Another result for Besag's original pseudo-likelihood is given by Guyon (1995). In this result stronger conditions are assumed than in the result of Comets and Janzura. The conditions assumed by Guyon lead to maximum pseudo-likelihood estimators such that

$$\hat{\boldsymbol{\theta}}_C^{(m)} \;\; AN\ (\mathbf{0}, \Sigma), \tag{9.36}$$

where

$$\Sigma^{-1} = H(\boldsymbol{\theta})\, J^{-1}(\boldsymbol{\theta})\, H(\boldsymbol{\theta}),$$

with these matrices defined as for composite likelihood in the previous section. Note, however, that evaluation of $H(\boldsymbol{\theta})$ and $J(\boldsymbol{\theta})$ may prove difficult.

9.3.3 The Importance of Asymptotic Results

One could argue that difficulties in applying asymptotic results, primarily in computing relevant covariance matrices, have made the topic of asymptotic inference for composite likelihood largely one of greater importance for statistical development than for application, at least for problems involving the asymptotic context of a single multivariate observation that expands in dimension. Experience with composite likelihoods even in the simpler situation in which replicate observations can be obtained of the same multivariate distribution seems to suggest that direct computation of sample versions of covariances may not be immediate in these cases either. Certainly for problems that would fall under the heading of Asymptotic Context II in these notes, the majority of applications have relied on the production of interval

estimates of parameter values through the use of parametric bootstrap procedures. If this is the case, one may wonder why I have devoted so much space to a discussion of asymptotic results for composite likelihood estimation. The basic reason is that the existence of asymptotic results provides some necessary justification to the use of parametric bootstrap procedures. Parametric bootstrap methods are based on simulation with the objective of learning about the sampling distribution of an estimator for the finite sample size that exists in a given problem (through the sampling distribution of some function of the estimator and the true parameter). In order for this to be meaningful we need some assurance that such a distribution exists. Any finite collection of values will give a set of relative frequencies that looks like a distribution, regardless of whether those values have been produced from a mechanism that approximates a true distribution or not. If we assume that our simulation procedure is approximating a distribution when in fact it is not, inferences drawn on the basis of the empirical relative frequencies in the set of simulated values are meaningless and potentially seriously misleading, even if we are not aware of the fact because our simulated values look like they are describing a distribution. That this is more than just a superfluous technical fine point may be seen by considering a simulation intended to illustrate the basic central limit theorem as taught in statistical methods courses. If multiple samples are simulated from a Cauchy distribution (that has no expected value or variance), centered by the location parameter (mode or median) and scaled by the square root of the sample variances divided by the square root of sample size, the resulting collection of values will describe an empirical distribution, but an empirical distribution that is approximating nothing. It is true that the empirical distribution will be quite unattractive in shape, but if we did not know what was supposed to happen we might

naively assume that this empirical distribution is approximating the sampling distribution of standardized sample means. Any inferential quantities formed on the basis of this empirical distribution would be quite misleading indeed. The same danger exists in settings with much greater complexity, and our ability to intuitively or empirically detect when things have gone awry is severely limited. Thus, the existence of asymptotic results that apply to estimators are necessary to lend reasonable assurance that a simulated empirical distribution is, in fact, approximating a distribution that actually exists and may be used to construct inferential quantities. A result that leads to a limiting distribution implies that there exists a sequence of distributions for finite sample sizes, and it is one of these that an empirical distribution of simulated values is then approximating.